



**Politécnico
de Tomar**

Escola Superior de Tecnologia
de Tomar

Checkpoint 2

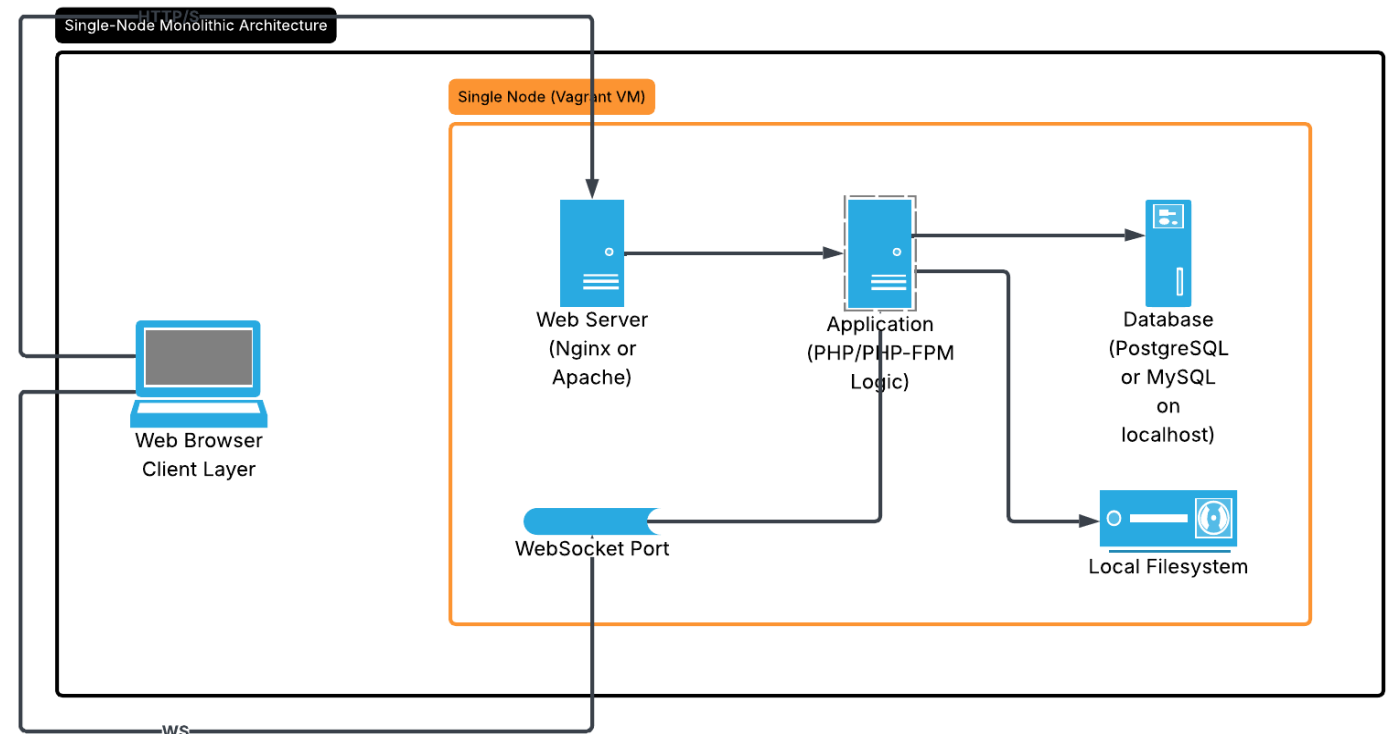
Final Project (A+B) - 2026

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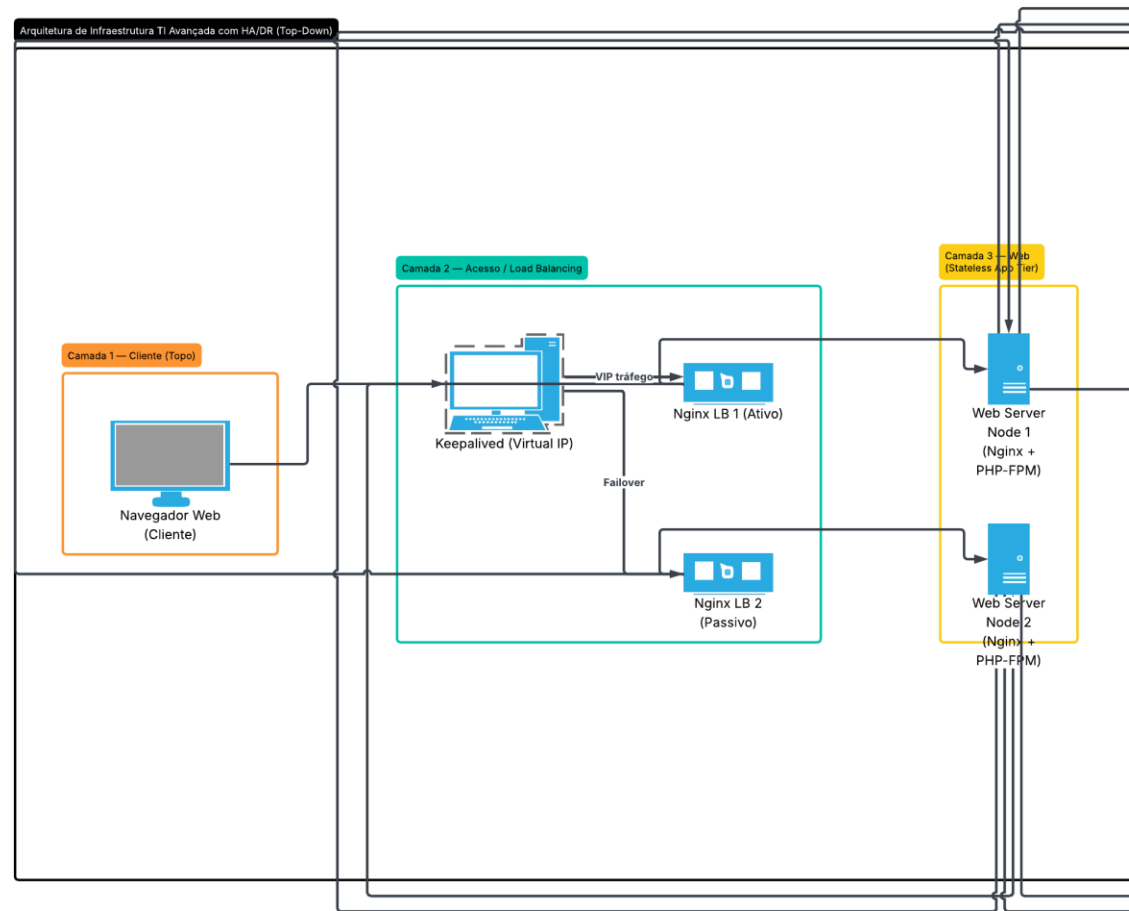
Architecture Diagram of Project O

Key Points :

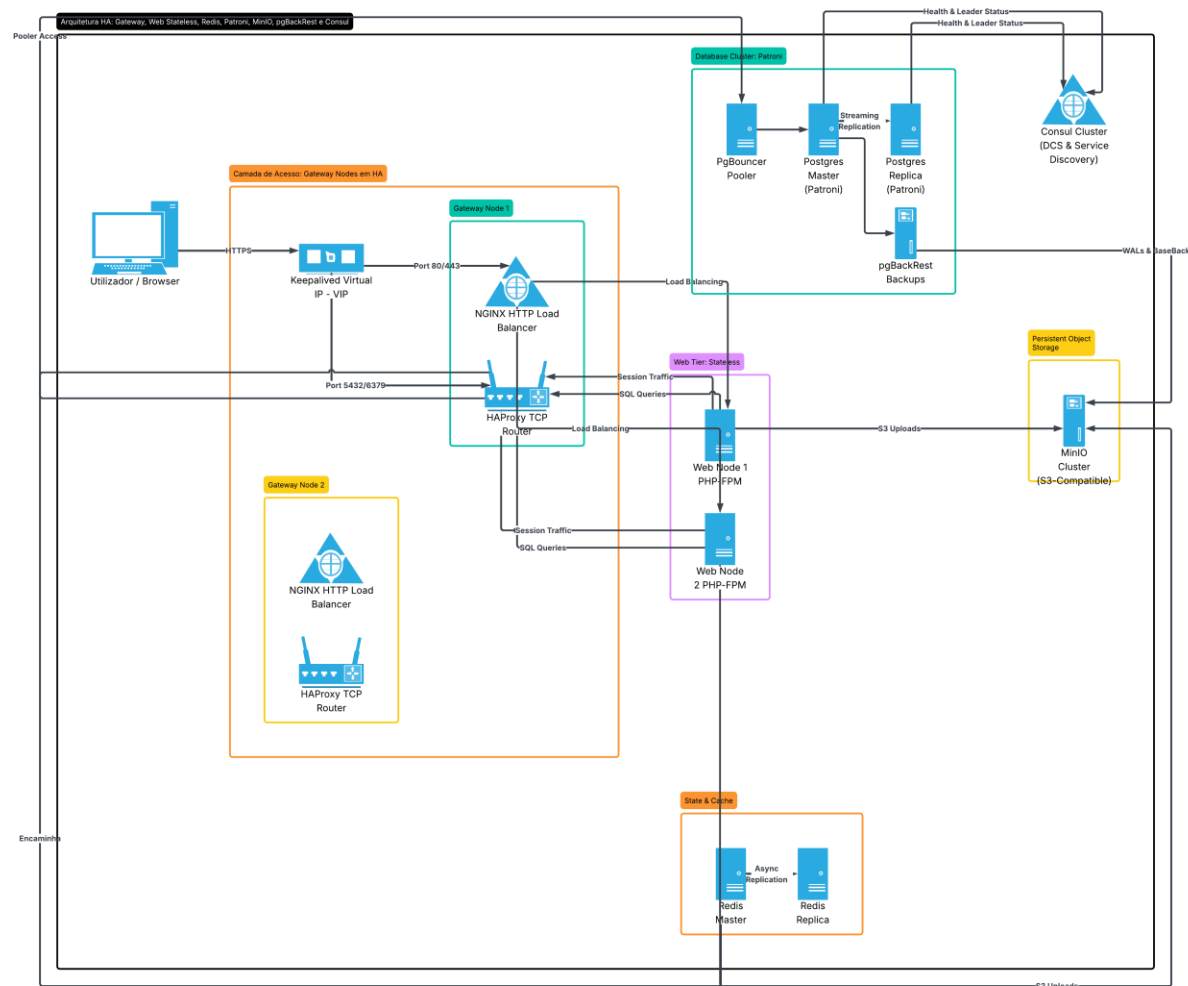
- **Single Point of Failure (SPOF):** Complete system downtime occurs if the underlying VM crashes.
- **Zero Horizontal Scalability:** Bound strictly by the hardware limits (CPU/RAM) of the single machine.
- **Trapped Local State:** User sessions and file uploads are locked to the local disk, preventing distribution.
- **Resource Contention:** Database, Web Server, and App logic compete directly for the same resources.
- **Coupled WebSockets:** Persistent connections are bound to a single local instance port.



Possible solution



Possible solution



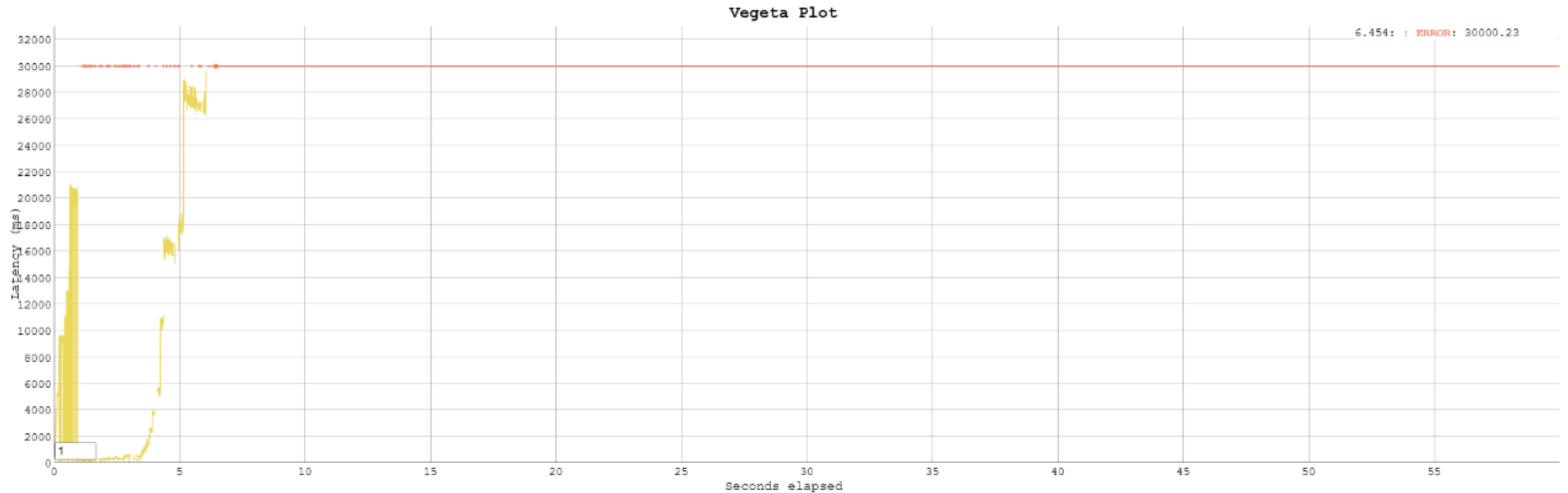
Testes Vegeta (Project 0)

```
Requests    ==[total, rate, throughput]    12000, 200.02, 7.90
Duration    ==[total, attack, wait]         1m30s, 59.995s, 30.001s
Latencies   ==[min, mean, 50, 90, 95, 99, max] 146.137ms, 28.908s, 30.001s, 30.001s,
30.001s, 30.001s, 30.021s
Success     ==[ratio]                      5.92%
Status Codes [code:count]                 |0:11289 200:711
Error Set:
Get "http://192.168.44.10/db.php": context deadline exceeded (Client.Timeout exceeded
while awaiting headers)
Get "http://192.168.44.10/db.php": dial tcp 0.0.0.0:0->192.168.44.10:80: i/o timeout
(Client.Timeout exceeded while awaiting headers)
```

```
[Thu Jun 04 21:36:33.417491 2026] [mpm_prefork:error] [pid 13373] AH00161: server
reached MaxRequestWorkers setting, consider raising the MaxRequestWorkers setting
```

echo "GET http://192.168.44.10/db.php" | vegeta attack -rate=200 -duration=60s -timeout=30s | vegeta report

Testes Vegeta (Project O)



Topologia do Cluster

Load Balancers (LB1, LB2)

Responsáveis pela entrada de tráfego e gestão da Alta Disponibilidade. Executam Keepalived, Nginx e HAProxy.

IPs: .11 e .12

Web/DB Servers (Web1, Web2)

Nós de processamento que alojam o servidor Apache, PHP e os nós PostgreSQL para armazenamento de dados.

IPs: .21 e .22

Tecnologias Core



Keepalived

Gestão do IP Virtual (.10) e failover automático entre balanceadores.



Nginx

Load Balancing HTTP para a camada Web (Apache).



HAProxy

Proxy TCP dedicado para tráfego PostgreSQL (Porta 5000).

HAproxy VS Nginx

Feature / Capability 	Nginx	HAProxy
Primary Design	Web Server + Reverse Proxy	Dedicated Load Balancer
Static Content	Excellent: Built-in caching for static files.	Poor: Not designed to serve files.
Advanced Health Checks	Premium: Needs paid Nginx Plus.	Native: Included in the open-source version.
Observability	Relies on logging; requires modules for metrics.	Native: Built-in Prometheus exporter & stats.

Automação com Ansible

Toda a infraestrutura é provisionada como código (IaC),
garantindo idempotência e rapidez no deployment.

- ✓ Roles modulares para cada serviço.
- ✓ Pipelining SSH para performance.
- ✓ Gestão automatizada de segredos.

```
--  
- hosts: all  
  become: yes  
  roles:  
    - common  
  
- hosts: loadbalancers  
  become: yes  
  roles:  
    - keepalived  
    - nginx  
    - haproxy  
  
- hosts: webservers  
  become: yes  
  roles:  
    - webserver  
    - postgres
```

Alta Disponibilidade

.10

IP Virtual (VIP)

O Ponto Único de Entrada



Através do protocolo VRRP, o IP .10 flutua entre o LB1 e o LB2. Em caso de falha do Master, o Backup assume o tráfego em menos de 2 segundos.



Isso garante que o utilizador final nunca perde o acesso ao "Projecto O".

Análise de Rutura (Vegeta)

```
echo "GET http://192.168.44.10/" | vegeta attack -duration=60s -rate=200 | tee results.bin | vegeta report
```

Requests	[total, rate, throughput]	12000, 200.02, 33.69
Duration	[total, attack, wait]	1m4s, 59.994s, 4.326s
Latencies	[min, mean, 50, 90, 95, 99, max]	2.4ms, 2.044s, 67.825ms, 10.949s, 12.729s, 16.195s, 30.002s
Bytes In	[total, mean]	8752513, 729.38
Bytes Out	[total, mean]	0, 0.00
Success	[ratio]	18.06%
Status Codes	[code:count]	0:9833 200:2167

DADOS CRÍTICOS

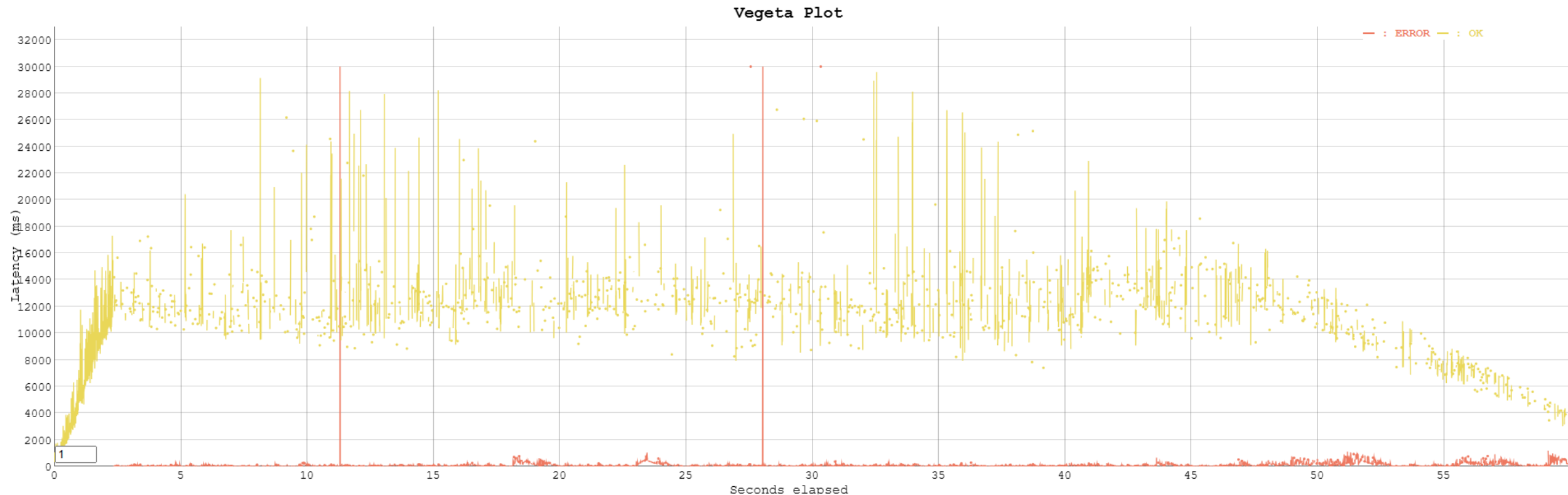
Status 0: 9833 falhas de ligação (Timeout/Refusal).

p99: 16.19 segundos (Saturação severa).

Throughput Real: ~33 RPS estáveis.

Conclusão Técnica: O ponto de rutura foi atingido de forma severa ao exigirmos 200 RPS durante 60 segundos. O limite físico de I/O de disco e CPU do computador host esgotou-se, obrigando o cluster a estabilizar a sua capacidade real em cerca de 33 pedidos por segundo.

Análise de Rutura (Vegeta)





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